FoMo

Forking Moment in Generative Trajectory as a Perceptual Distance

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Project page
jhlew.github.io/FoMo

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pip install fomo-iqa
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DSAIL Data Science & AI Laboratory

NEURAL INFORMATION PROCESSING SYSTEMS

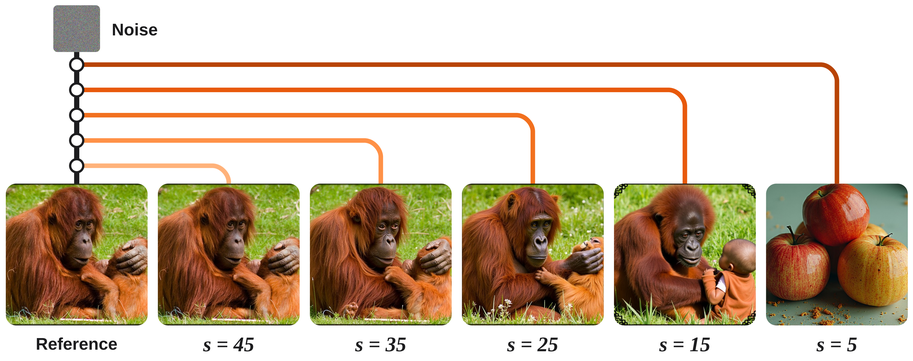
The problem
Human labels are scarce.

KADID-10K
10,125 images, 30 ratings each

480k
FoMo pairs

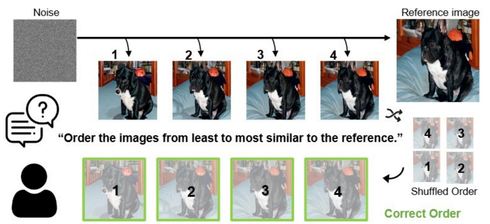
0 human ratings

The idea: automated labeling with divergence points in generation process
Fork early: far apart. Fork late: close.



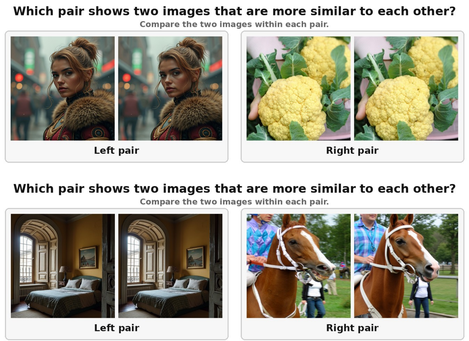
Does it match people?
People agree, in two studies.

Same reference



0.970
rank correlation · 1,982 annotations

Across references

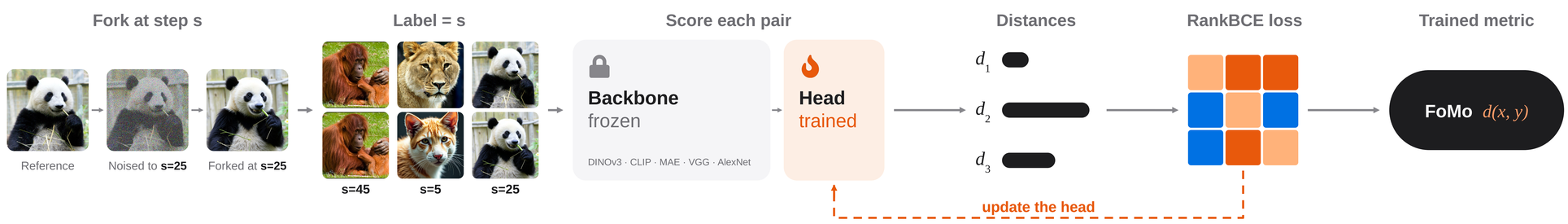


90.2%
of 5,713 annotations

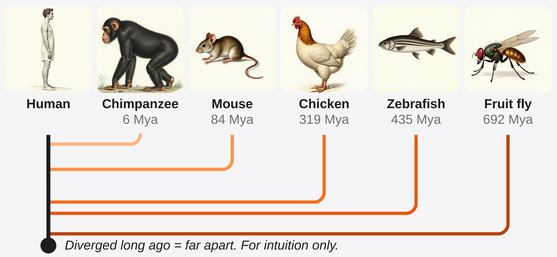
92.8%
of items, majority

How to train
Forking moments are the labels.

A small head on a frozen backbone learns to rank pairs the way the forks do. No human in the loop.



Side note
An unofficial, conceptual analogy

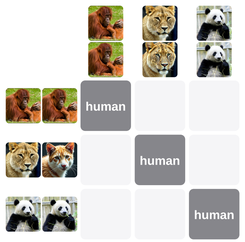


Noise ≈ common ancestor **Fork step** ≈ divergence time

Times: TimeTree, square-root scale. Illustrations: FLUX.1-dev.

Why RankBCE
Rank every pair, not one.

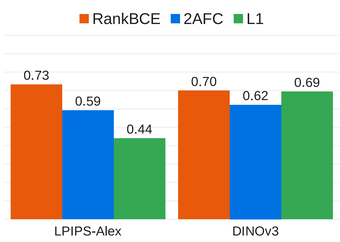
64 2AFC comparisons



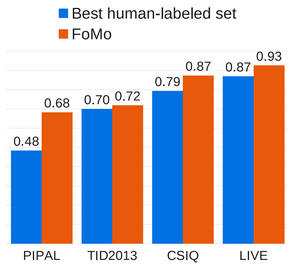
4,096 RankBCE comparisons



Same labels, PIPAL SROCC



Results
Ahead of human labels on average.



SROCC, FoMo minus best human-labeled set

	Alex	VGG	DISTS	DINOv3	CLIP	MAE	DreamSim
PIPAL	+0.111	+0.048	+0.051	+0.374	+0.332	+0.294	+0.016
TID2013	−0.008	−0.094	−0.143	+0.100	+0.217	+0.048	−0.014
CSIQ	−0.007	−0.037	−0.020	+0.195	+0.297	+0.097	−0.018
LIVE	−0.004	−0.006	+0.004	+0.132	+0.098	+0.110	+0.003

FoMo ahead · Human labels ahead

7 / 7
PIPAL backbones

12 / 12
frozen-ViT cells

18 / 28
all cells